

Zain Munir

SOFTWARE ARCHITECT

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Open to full-time and freelance roles. Gulf relocation (UAE / KSA) welcome; comfortable remote worldwide.

PROFESSIONAL SUMMARY

Software Architect and backend engineer with 6+ years designing and shipping scalable, secure distributed systems. I decompose monoliths into clean microservices, design event-driven and CQRS architectures, and own the hard technical calls behind products used daily by tens of thousands of people. Strong in Node.js and NestJS, TypeScript and Python, with hands-on delivery of payment platforms, ERP integrations, zero-data-loss migrations, and production-grade observability. Increasingly building AI into products with the OpenAI API, RAG and embeddings.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript (ES6+), Python, SQL

Architecture: Microservices, Event-Driven Architecture, CQRS, Enterprise Integration Patterns, Middleware, Domain Boundaries, System Design

Backend and APIs: Node.js, NestJS, Express, FastAPI, REST, GraphQL, Swagger / OpenAPI

Messaging and Async: Apache Kafka, RabbitMQ, Bull Queue, Redis, Pusher (real-time)

Data: MongoDB, PostgreSQL, MySQL, Redis, Firebase

Cloud and DevOps: AWS (EC2, S3, RDS, Copilot), Docker, CI/CD, PM2, Linux

Security: JWT, OAuth2, RBAC, Helmet, Rate Limiting, Input Validation, XSS and NoSQL Injection Prevention

Observability: New Relic, OpenTelemetry, Winston (structured logging), distributed tracing

AI: OpenAI API, RAG, Embeddings, Prompt Engineering, LLM-powered features, AI-assisted development

Practices: Code Review, Mentoring, Agile / Scrum, TDD

PROFESSIONAL EXPERIENCE

Software Architect / Senior Backend Engineer

SimpliEd, Karachi, Pakistan

Jan 2023 - Present

- Set the technical direction and coding standards for an EdTech platform now serving 25,000+ daily users across multiple schools; lead architecture and code review for the backend team.
- Decomposed a tightly coupled monolith into independent NestJS microservices behind a secure middleware layer (JWT, OAuth2, RBAC, request validation, centralized error handling), improving scalability, deployability, and team autonomy.
- Architected a universal payment-receiving microservice that lets any partner bank integrate online payments with minimal configuration, plus a hardened webhook-receiver service (rate limiting, Redis caching, Helmet) for inbound payment callbacks.
- Built a standalone async job-processing service with 40+ Bull Queue processors (invoice and result-card PDF generation, push notifications, SMS, automated attendance, admission workflows), fully decoupling async workloads from the API.
- Documented 100+ REST endpoints with Swagger / OpenAPI and maintained versioned API contracts, accelerating mobile and third-party integration.
- Executed a full migration from legacy systems to the new platform with zero data loss.
- Established production observability with PM2, Winston (daily log rotation), and New Relic APM with OpenTelemetry distributed tracing; led root-cause analysis to maintain high availability.

Backend Software Engineer

Alpha Mu Digital Ventures, Karachi, Pakistan

Sep 2022 - Dec 2022

- Built asynchronous web-scraping APIs in Python and FastAPI with concurrency and rate limiting to aggregate product data at scale across multiple e-commerce platforms.
- Delivered a data-migration microservice moving records from Firebase to PostgreSQL with full data integrity and referential consistency; designed normalized schemas with efficient indexing.
- Containerized and deployed services to AWS with Copilot, and cut response times under load through query refactoring and Redis caching.

Backend Software Engineer

Queno, Karachi, Pakistan

Dec 2019 - Aug 2022

- Led the decomposition of a monolithic application into microservices, substantially improving scalability and independent deployability.
- Integrated ForeePay and Blinq payment gateways and the QuickBooks ERP for automated financial record-keeping; built a finance service generating real-time balance sheets and income statements.
- Designed and managed schemas across MongoDB and PostgreSQL for financial, student, and payment models under production load, with RBAC across student, parent, and admin roles.

SELECTED PROJECTS

Shipwright - Autonomous AI software company: role-based AI agents take a ticket through spec, build, code review, QA, and ship. FastAPI, LangGraph, PostgreSQL / pgvector, Next.js. github.com/iamzainmunir/shipwright

DevPulse - Slack-driven, read-only pull-request and sprint tracker with multi-provider parsing, sprint metrics, and developer leaderboards. Next.js 15, TypeScript, Drizzle, PostgreSQL. github.com/iamzainmunir/pr-tracker

StockWise - Offline-first, multi-currency inventory and business platform for SMBs: multi-branch inventory, customer and vendor ledgers, payroll, and invoicing. Next.js, Prisma, PostgreSQL, PWA.

github.com/iamzainmunir/stockwise

EDUCATION

M.S. in Computer Science and IT (MSCSIT)

2021 - 2023

NED University of Engineering and Technology, Karachi

B.S. in Computer Science (BSCS)

2016 - 2019

University of Karachi (UBIT), Karachi

LANGUAGES

English (Professional), Urdu (Native)